Commissioning & test plan (concept), with genset and MV switchgear selection points

Header	
Title	Commissioning & test plan (concept) for the fictional 30 MW IT campus, plus genset and MV switchgear selection points
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Status	**CONCEPT / DRAFT — requires review by a licensed electrical engineer**
Scope	The campus of Projects 01–04 (Dalarna SE3, 6 halls × 5 MW IT), with hall DH1 as the worked example. It is part 2 of Project 05 (research and concept study). A short section covers what changes if the same campus were built in Finland.
Assumptions	All ratings come from DB-001 Rev P2 and CALC-003/004 and are not changed here. Commissioning "levels" are industry practice, not one standard [Inference]. The test durations and acceptance values below are my proposals [Inference] unless a source is given. Nothing here describes work I have done or signed for. I hold no Swedish elinstallatör authorisation, and design and installation responsibility stays with the authorised parties.
Codes (pointers)	IEC 60364 / SS 436 40 00 · SS-EN 61936-1 · EN 50600-2-2 · LVD 2014/35/EU · IEC 62271-200 · IEC 61439 · IEC 62040-3 · ISO 8528-1/-5 · IEC 61850: all [Unverified] unless opened. EU F-gas Regulation 2024/573 Art. 13(9) [Verified].

Labels: [Verified: opened source] · [Vendor claim] · [Search-level] · [Unverified] · [Inference]. Retrieval date for all sources is 2026-10-29. Regulation and grid-code items carry their assumption and stay [Unverified] unless the page was opened.

1. Commissioning levels used here

There is no single standard that defines "L1–L5". Vendors and consultants use similar but not identical lists [Search-level: HIOKI, Avtron, Aggreko, constructandcommission, sources 9–12]. The structure below is a common form; the campus content in each row is mine [Inference].

Level	Purpose	Campus content (DH1 example)	Evidence produced
L0 Design review	Check that the design can be tested	Load-bank points on every MSB (DB-001 §7.4); test switches and CT test blocks in relays; IEC 61850 test mode; a way to load-test the 20 kV feeders	Commented design, test-point list
L1 FAT	Factory acceptance test	MSG-20 and HV-DH1: routine tests, interlock check, relay settings loaded. UPS 2 × 1 000 kW: IEC 62040-3 type and routine reports [Unverified], step-load and transfer test at the factory. Gensets: load acceptance steps, Isc ≥ 300 % for 2 s (DB-001), −30 °C rating evidence (DB-001 design range)	FAT reports, witnessed
L2 Delivery / installation	Receipt and installation checks	Transport damage, torque marking on busways, cable test (VLF / tan δ on 3×1×240 Al 20 kV feeders [Inference]), IR and continuity on LV	Installation checklists
L3 Start-up / pre-functional	Energise each item alone	Primary and secondary injection, CT polarity, 87T stability, 67N direction on the resonant-earthed 20 kV network, Petersen coil tuning, battery string checks	Energisation records
L4 Functional	Each system under its own load	Genset on load bank via Q-LB; UPS on load bank (rectifier walk-in, battery discharge); ATS IL-3 open-transition timing; EPMS point-to-point	Functional test scripts signed
L5 IST	All systems together under heat load	Utility loss, genset start, ATS and UPS transfer, redundancy and recovery while white space carries load-bank heat (see §3)	IST report, punch list

2. Load-bank plan and one constraint I found in our own design

- Each MSB has a load-bank connection, Q-LB 2 500 A (DB-001 §7.4, SLD-003).
- At 400 V, 2 500 A is √3 × 0.4 kV × 2.5 kA ≈ 1.73 MVA. The IT genset is 2 750 kVA / 2 200 kWe. Through Q-LB alone, a genset can therefore be tested to about 1.73 MW at p.f. 1. That is 79 % of kWe and 63 % of kVA [Inference, arithmetic].
- The N-1 design point of a block is 1.67 MW (DB-001), which is 96 % of what Q-LB can carry. The **block design load can be proven through Q-LB, but the genset's full rating cannot**. The genset's own N-1 design load, including UPS losses and recharge, is ≈ 1.93 MW (CALC-002), which is also above Q-LB [Inference, arithmetic]. DB-001 §7.4 records this as an open point.

- Options: a second load-bank point, a temporary connection on the genset side of Q-GI, or full-rating tests at the factory only (L1) [Inference]. I would raise this at L0.
- White-space heat load for L5: 5 MW per hall ≈ 200 rack-mounted load banks at 25 kW, or AI-row units at 80–130 kW per rack position. They are fed through the normal busways and tap-offs so the real path is tested [Inference].
- Cummins gives a rule of thumb that a genset should run above 30 % of its rating to limit wet stacking [Vendor claim, source 4]. That supports monthly testing on the load bank or on the building load rather than at no load.

3. Integrated systems test (L5), DH1 scenarios

All scenarios run with the hall at its stated heat load from load banks. Acceptance values are proposals [Inference] unless a document is named.

#	Scenario	Expected behaviour (from the design)	Pass check
1	Loss of one 130 kV line	H-arrangement keeps both 63 MVA transformers fed; no 20 kV interruption	No UPS on battery
2	Loss of T1 (130/20 kV)	20 kV coupler auto-closes only after the T1 incomer is proven open. Break-before-make, **2-of-3 logic: at most two of the incomer A, incomer B and coupler closed**. T1//T2 never parallel (DB-001 §7.2)	Interlock blocks a third close; UPS rides through
3	Total grid loss	All block UPSs on battery; G-A to G-D start; on load ≈ 10–15 s (CALC-004); ATS transfers by open transition (IL-3)	IT load not interrupted; UPS battery time used < 5 min
4	One genset fails to start	Under 4N/3 the other three blocks carry 5 MW at 1.67 MW each, 83 % of UPS and 88 % of the genset (DB-001). The BESS is not counted (CALC-004)	No pair-group lost
5	Block failure (UPS output fault on block A)	Pair-groups AB, AC and AD shift to B, C and D	Busway and STS/ATS loading ≤ design
6	Retransfer to grid	Open transition only. Closed transition is blocked because it would give 81.8 kA against a 65 kA rating (DB-001)	Interlock proven by attempting a closed-transition command
7	UPS battery discharge	Run on battery at block load until the low-voltage alarm; compare with 5 min at Pr EoL / 8.1 min in N-1 (DB-001). A new battery gives a longer time; correct for state of health [Inference]	Measured time ≥ the design value corrected to BoL
8	MBB / SBS maintenance transfer	Key-interlocked wrap-around bypass (DB-001 §7.4)	Key sequence cannot be broken
9	Arc-flash protection trip	Optical arc relay on MSB-A trips S1-Q1A (20 kV). ZSI and ARMS function (DB-001 §7.4)	Trip by test light and primary-injection current check
10	Fuel transfer under genset load	Day tank refill while running; pump power path concurrently maintainable [Verified, Uptime source 1]	No genset derate
11	BESS island / grid events (optional)	Not part of the Tier III design (CALC-004)	Only after a separate island protection study

4. Genset selection points (campus-specific)

- **Rating basis.** DB-001 uses 2 750 kVA / 2 200 kWe **DCC**, a manufacturer's data-centre continuous rating.
- Uptime's Tier III/IV requirement is that engine generators carry the design load with no limit on runtime; Uptime treats the utility as an "economic alternative". These are quoted via a Cummins white paper, so they are [Vendor claim] until the Uptime original is opened (source 4).
- The same paper says Uptime accepts a manufacturer's certification of unlimited duration, including a site-specific rating letter [Vendor claim].
- **ISO 8528-1 rating words** (ESP, PRP, COP, LTP and DCP from 2018) [Search-level, sources 4–5].
- Cummins states that ISO **DCP** is "not appropriate" for Uptime Tier III/IV, because its definition references a reliable utility [Vendor claim].
- **Load steps.**
- IT gensets see the UPS rectifier walk-in, not a hard step. Rectifier ramp time and the ISO 8528-5 performance class should be in the specification [Unverified].
- Mechanical gensets are governed by the 400 kW chiller step ≤ 33 % (DB-001).
- **Short-circuit support.** Sustained Isc ≥ 300 % for 2 s (PMG or AREP) so the genset ACB, with Isd ≤ 2.5 × Ir, and downstream devices trip (DB-001). Verify at FAT with an alternator test report.

- **Fuel.**
- HVO100 for 24 h at design load, ≈ 51 m³ gross per hall (DB-001).
- Uptime sets at least 12 h at N for all Tiers. Stored fuel counts only after removing the redundant tank [Verified, source 1].
- The engine maker's HVO approval is needed in writing (fuel standard reference [Unverified]).
- **Climate.** Gensets are rated −30 … +35 °C for a −28 °C winter design (DB-001). The cold-start and low-load behaviour evidence should come from the vendor; I found no source on cold-climate commissioning tests and do not add one.
- **Paralleling.** Not used: one genset per block, no paralleling bus, open-transition ATS only (DB-001).

5. MV switchgear selection points (MSG-20 and HV-DH1)

- **Ratings** as in DB-001: MSG-20 24 kV, 2 500 A, 25 kA / 3 s, Ip 62.5 kA. HV-DH1 24 kV, 1 250 A, 25 kA / 1 s.
- **Classification** per IEC 62271-200 as written in DB-001: LSC2B-PM and IAC AFLR 25 kA 1 s.
- The IAC code gives the accessibility type (A = authorised personnel), the accessible sides (F front, L lateral, R rear), the test current and the duration [Search-level, Schneider FAQ, source 7].
- IAC is a type test of the switchgear. Room pressure rise and gas-exhaust routing are not covered by it and need a separate building calculation [Search-level, Roxtec, source 8].
- The edition of IEC 62271-200 to cite is [Unverified].
- **Insulating medium.**
- **EU F-gas Regulation 2024/573 Art. 13(9)(a) prohibits putting into operation MV switchgear for primary and secondary distribution up to and including 24 kV that uses F-gases in its insulating or breaking medium from 1 January 2026** [Verified, EC FAQ, source 6]. Derogations exist, for example orders placed before 11 March 2024 and procurement-based exceptions.
- On this basis DB-001 §7.2 requires MSG-20 to be SF_6-free; any derogation would have to be documented [Inference].
- The 130 kV side falls in the > 52–145 kV band: only GWP < 1 from 1 January 2028 [Verified, source 6].
- **Earthing system.** The 20 kV network is resonant earthed (Petersen coil ≥ 50 A plus a 10 A resistor), so feeder relays need admittance-based or transient 67N (DB-001). The specification should require the relay vendor's proof for compensated networks and a staged earth-fault test at L3/L5 [Inference].
- **Arc protection.** Optical arc plus reverse-blocking busbar protection; MSG-20 incident energy falls from about 106 to 11 cal/cm² (DB-001 / CALC-003). Test light sensors and the trip path at L3.
- **Interlocks.** Incomer, coupler and incomer 2-of-3 logic with break-before-make (DB-001 §7.2). Check it mechanically (key or electrical) and in the IEC 61850 logic, and test both at FAT and IST.
- **Monitoring.** IEC 61850 at MV; 0.2S revenue meters at the incomers and BESS (DB-001). A GOOSE test plan should be in the FAT.

6. If the same campus were in Finland (Muhos-type site)

The assumption here is a transmission-connected (Fingrid) demand facility above 30 MW. All points are pointers for the day-one review, not legal advice.

- **Grid code.**
- KJV2018 is Fingrid's current demand-connection specification, based on EU 2016/1388 [Search-level, source 13].
- The **KJV2026** draft was out for consultation from 12 June to 21 August 2026. It sets requirements for demand facilities above 30 MW, explicitly including data centres: fault ride-through and the capability to limit power intake. The target for entry into force is March 2027 [Verified, source 14].
- A separate Fingrid document, "Technical requirements for large demand facilities" (12 June 2026), **applies immediately** to facilities connected directly to Fingrid's transmission grid [Verified, source 14].
- Effect on this plan: the L5 IST and the grid-compliance tests would need a fault ride-through verification path. That means simulation with Fingrid-format PSS/E or PSCAD models (Fingrid modelling instruction for data centres, source 15 [Search-level]) plus site tests of the power-limiting function [Inference].
- **Installation rules.**
- The Electrical Safety Act 1135/2016 §§ 31–33: installations meet the essential requirements if they follow the standards listed by the safety authority [Search-level, source 16].
- Tukes lists the SFS 6000 series for installations up to 1 000 V AC and SFS 6001 above 1 000 V [Search-level, source 17].
- SFS 6000 covers up to 1 500 V DC, so an 800 VDC system would sit in the LV series [Search-level, SFS store page, source 18].
- A commissioning inspection and record are required before an installation is put into use [Search-level, source 16]. The Finnish names and the scope of that inspection are [Unverified].
- **Switchgear.** The EU F-gas rule above applies in Finland as in Sweden [Verified, source 6; EU regulation].

7. Questions I would ask on day one

1. Which commissioning framework and level names does the owner use, and who signs each level?
2. Is a full-rating genset load test required on site, or is the factory test accepted? This matters because of the Q-LB 2 500 A limit in §2.
3. What genset rating wording does the owner accept (manufacturer unlimited-hours, ISO COP, ISO DCP)? Is there a site rating letter?
4. Is the IST heat load rack-mounted load banks through the real busways, or temporary feeds?
5. Which MV switchgear medium (vacuum + dry air, or other) is on the approved vendor list, and is any F-gas derogation planned?
6. Which grid code applies to the connection point (Svk / DSO in Sweden; Fingrid KJV2018, the large-demand requirements, or KJV2026 in Finland), and who owns the ride-through study?
7. Is a staged 20 kV earth-fault test on the resonant-earthed network allowed by the network owner?
8. How is the UPS battery runtime test corrected for a new battery against the EoL design value?
9. Which relay test sets and IEC 61850 tools are used, and is there a GOOSE simulation step before IST?
10. Who is the authorised electrical installer / supervisor of record for each phase?

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Fictional case · D. Wu portfolio · no employer, client or vendor data · CONCEPT / DRAFT — requires review by a licensed electrical engineer.